Welcome

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1. What is this course about?

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Why should you study algorithms?

- This course is about problem-solving and communication.
- Two pillars:
  - algorithm design is the task of designing a process which, when executed, solves a particular problem;
    - This process might be a way of calculating something, selecting something, etc.
  - algorithm analysis is the task of assessing such a process (either your own work or someone else's) with regard to the problem.
    - Does it always give the intended result?
    - How quickly does it run?

Algorithm analysis: why is correctness important?

- A program which fails a test case is wrong, but a program which passes a test case isn't <u>necessarily</u> correct.
- Failure on an obscure case can be catastrophic!
- Testing is good, good testing is better, proof is best.
- What makes a strong test?
- How do you prove (or at least justify <u>in general terms</u>) correctness and explain it to a peer?

Algorithm analysis: why is efficiency important?

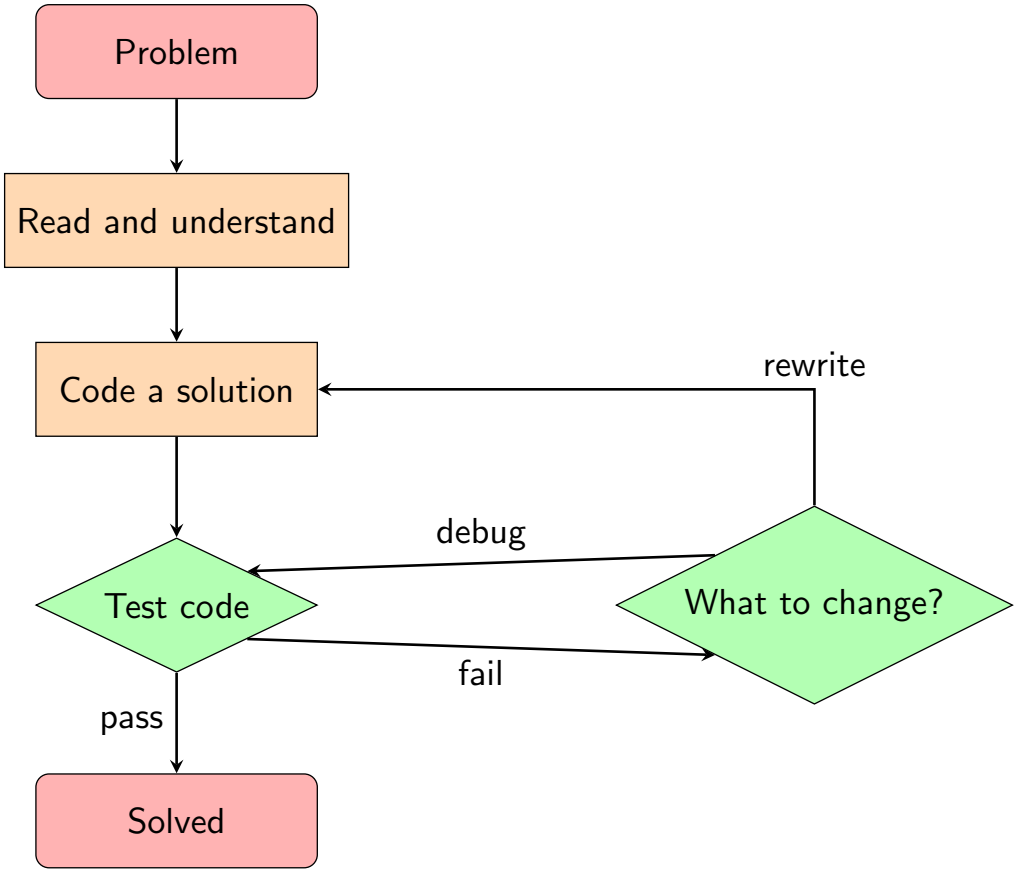
- Using better hardware might save a small (or even large) percentage of the running time of a program.
- However, a more efficient algorithm can make staggering improvements!
- Sorting one million numbers could take an hour using bubble sort, or a second using merge sort.
- For more sophisticated algorithms, details such as the choice of data structures are also important.

Why is algorithm design important?

- Algorithm design is the "planning stage", where we decide what our program should do.
- We have all tried to code programs only to find a fundamental flaw in our approach later on.
- A systematic approach to algorithm design separates engineers from "code monkeys".

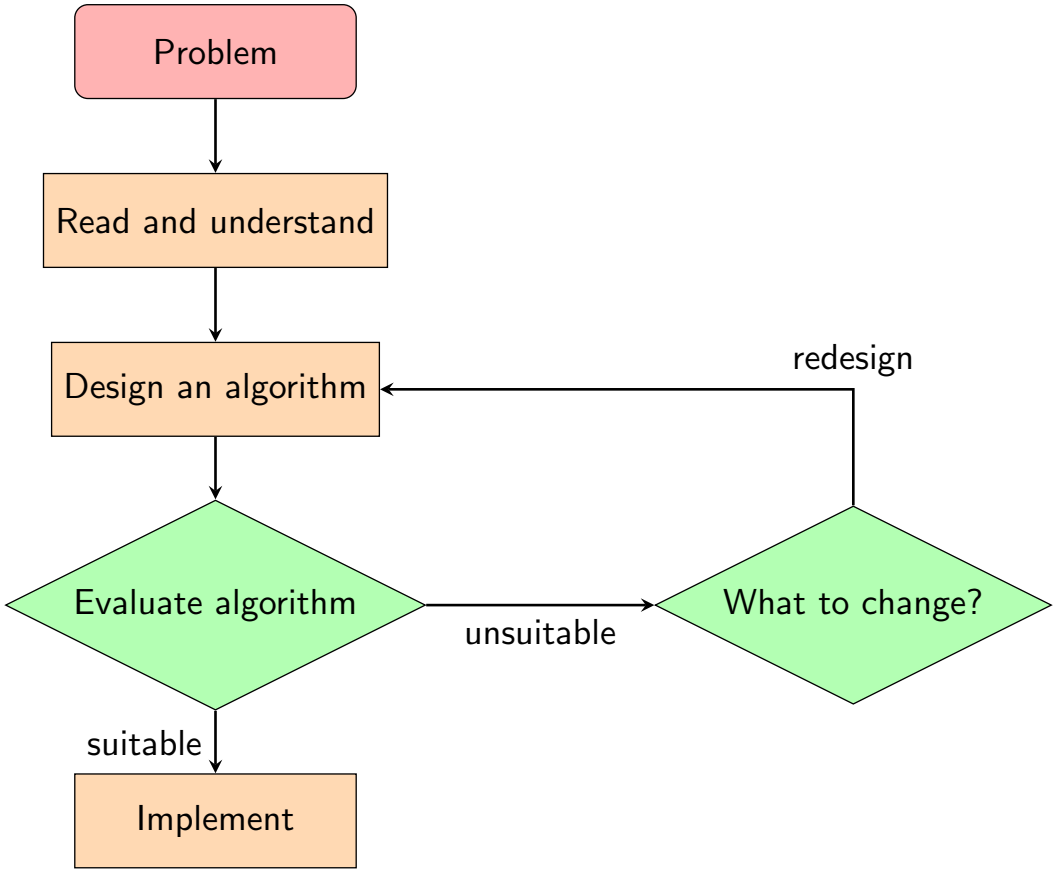
Code-first approach

Solving the problem as you code can lead you into a "debugging vortex", applying patches without fixing the underlying issue, or to accept incorrect code that passes weak tests.



Design-first approach

Separating design from implementation allows high-level reasoning to confirm that the algorithm meets the requirements of the problem before making implementation decisions.



Separating design from implementation

- We will talk about algorithms in plain English. We will avoid pseudocode, and we will especially avoid using code in any programming language.
- Implementation is outside the scope of this course.
- We'll focus on the big picture, to prepare you to:
  - work in teams on large projects
  - communicate clearly and persuasively with stakeholders including colleagues, clients and the public
  - compile design briefs and documentation,

  which are all important skills for the modern computing professional.

Course Learning Outcomes

1. Explain how standard design techniques are used to develop algorithms.
2. Solve problems by creatively applying algorithm design techniques.
3. Communicate algorithmic ideas at different levels of abstraction.
4. Evaluate the efficiency of algorithms and justify their correctness.
5. Apply the LaTeX typesetting system to produce high-quality technical documents.

What is an algorithm?

- An algorithm is a collection of precisely defined steps that can be executed mechanically, i.e. without intelligent decision-making.
- Designing a recipe involves creativity, executing it does not; the same is true of algorithms.
- The word "algorithm" comes by corruption of the name of Muhammad ibn Musa al-Khwarizmi, a Persian scientist 780–850, who wrote an important book on algebra, "Al-kitab al-mukhtasar fi hisab al-gabr wa'l-muqabala".

Scope

- In this course we will deal only with algorithms that are:
  - sequential: comprised by a sequence of steps, which can only be executed one at a time, and
  - deterministic: each step always gives the same result for the same input.
- Parallel algorithms and randomised algorithms are interesting and powerful, but beyond the scope of this course.

Problem structure

- Problems may have ‘flavour text’: a brief story for context. Abstracting this information away is a key first step.
- Algorithm design tasks may also include an input and output specification if it is not otherwise clear.
- In this course, we are usually not concerned with the format in which this information is presented, nor with the minute details of how an implementation of the algorithm accesses input or produces output.
- However it is crucially important that our algorithm <u>always</u> transforms valid input into the appropriate output <u>efficiently</u>.

Problem structure

- We'll usually measure the efficiency of algorithms using asymptotic analysis, rather than the exact time or number of operations taken.
- You will have already encountered big-Oh notation.
  - For example, a process consisting of two nested loops "for each i from 1 to n, for each j from 1 to n" takes $O(n^2)$ time.
  - We'll formalise these ideas in the next lecture.
- We'll usually use the unit-cost model of computation, where each machine operation ($+$, $-$, $\times$, $\div$ and accessing memory) takes constant time.
  - The only exception is when we discuss divide-and-conquer algorithms to multiply large integers: there we'll use the logarithmic-cost model, where these operations take time dependent on the width (number of bits) of the input numbers.

Example problem: Geese in a Trenchcoat

Problem

There is a gaggle of n geese, who have heights $h_1, \ldots, h_n$. Each h_i is a positive integer.

They need to select two of them to wear a trenchcoat together, with one standing on top of the other. The trenchcoat will only fit if the two geese have total height exactly equal to T.

Design an algorithm which runs in $O(n \log n)$ time and determines whether there are two geese who can wear the trenchcoat together.

Example problem: Geese in a Trenchcoat

Artwork credit: Larissa Yip

Abstraction

Design an algorithm which runs in $O(n \log n)$ time and, given an array h consisting of n positive integers and a positive integer T, determines whether there are two indices $i < j$ such that $H[i] + H[j] = T$.

- If we fix index i (one of the geese), then we know that a matching element must contribute the rest of the total (make up the rest of the required height).
- Let's try all values of i.

Attempt 1

For each goose i, check whether a matching goose (whose height is $T - h_i$) exists, and answer yes if one is found.

- This is not an algorithm!
- It does not describe how to actually perform this check.
- Why does this matter?
  - For a small number of geese, a human could follow this process, checking by eye. If there were say one million geese, it's not at all clear how to do the check.
  - Perhaps this check takes time?

Attempt 2

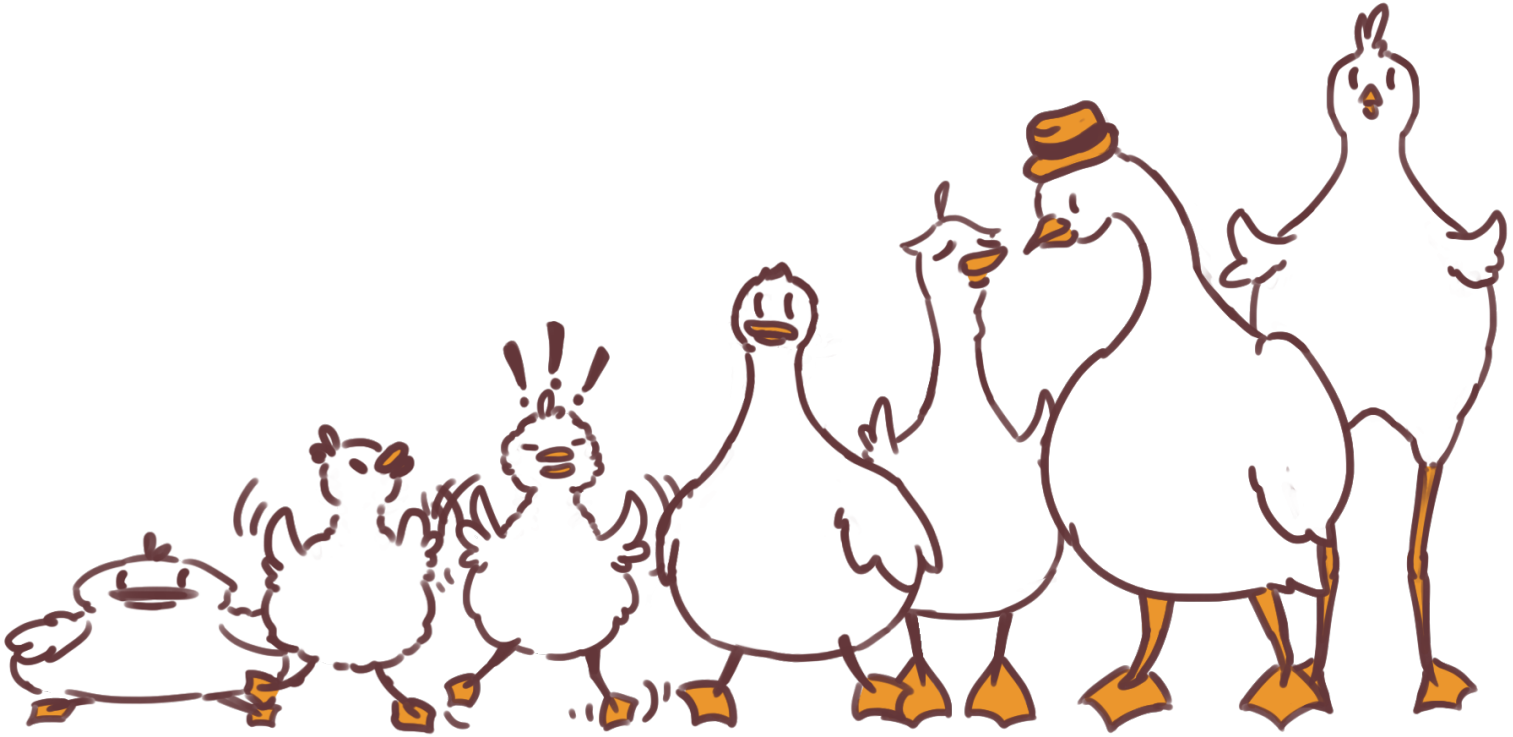
For each goose i, a matching goose must have height $T - h_i$. For each other goose j, check whether h_j equals this quantity, and answer yes if one is found.

- This is actually an algorithm.
- It is also a correct algorithm.
  - Does it correctly answer all 'yes' instances?
  - Does it correctly answer all 'no' instances?
- Unfortunately it's too slow: "for each i (of which there are n), for each j (of which there are n), do something" is already n^2 iterations.
- Can we search efficiently for the correct j, rather than trying them all? Perhaps if the heights are well organised.

Algorithm

Sort the geese in ascending order of height, using merge sort. For each goose i, a matching goose must have height $T - h_i$. Since the heights are now nondecreasing, we can search for this required length using binary search, and answer yes if it is found.

Example problem: Geese in a Trenchcoat



Artwork credit: Larissa Yip

Example problem: Geese in a Trenchcoat

- Is this algorithm correct?
  - Does it correctly answer all 'yes' instances?
  - Does it correctly answer all 'no' instances?
- What is the time complexity of this algorithm?
  - How long does the sort take?
  - How many times is the binary search run?
  - How long does each binary search take?
- You will usually have to justify that your algorithm is correct and that it runs within the specified time complexity.

Algorithm Design Rubric

- **Clarity**: the algorithm is communicated clearly, unambiguously and concisely. All terms used in the algorithm are well-defined. Where appropriate, clearly annotated diagrams supplement the algorithm description.

- **Style**: the algorithm is written in plain English and not written in (pseudo-)code.

Algorithm Analysis Rubric

- **Time complexity**: the time complexity is correctly measured, with clear reasoning and correct use of asymptotic notation. The time complexity analysis is internally consistent.

- **Correctness**: the proof is correct. The proof does not leave major gaps for the reader to complete. Computations are correct. The conclusion to the proof is present.

- **Clarity**: the proof is communicated clearly, unambiguously and concisely. All terms and notations used in the proof are well-defined. Where appropriate, clearly annotated diagrams supplement the proof.

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Modules

1. Foundations (week 1)
   - Foundations of algorithm design (data structures, sorting and searching)
   - Foundations of algorithm analysis (correctness, time complexity)
2. Greedy algorithms (weeks 2–3)
   - When local considerations align with global
   - Applications: file compression, graphs
3. Dynamic programming (weeks 3–5)
   - Recursion with <u>overlapping</u> subproblems
   - Applications: graphs, string matching

Modules

4. Divide and conquer (weeks 7–8)
   - Recursion with <u>independent</u> subproblems
   - Applications: big integer multiplication, signal processing
5. Flow networks (week 8–9)
   - Routing ‘flow’ through a graph
   - Applications: allocation and matching problems
6. Intractable problems (weeks 9–10)
   - Recognising, analysing and relating ‘hard’ problems

Recommended textbook

Kleinberg and Tardos: Algorithm Design
paperback edition available at UNSW Bookshop

- excellent: very readable textbook (and very pleasant to read!);
- not so good: as a reference manual for later use.

An alternative textbook

Cormen, Leiserson, Rivest and Stein: Introduction to Algorithms
4th edition now available at UNSW Bookshop, 3rd edition also useful

- excellent: to be used later as a reference manual;
- not so good: somewhat formalistic and dry.

Prerequisites

- COMP2521 or COMP9024
  - Fundamental data structures
  - Fundamental algorithms
- MATH1081 or COMP9020
  - Proofs
  - Written communication skills

Extended courses

- The extended courses COMP3821/9801 run once a year.
- Differences in content and assessment

Courses to consider

- COMP4121 Advanced Algorithms
  - Real-world algorithms, including randomisation (and therefore probability and statistics)
- COMP4128 Programming Challenges
  - Algorithms and C++ implementation for small, self-contained problems, as in contests.
- COMP2111 System Modelling and Design, COMP3153 Algorithmic Verification
  - Formal and rigorous methods for verifying software
- COMP4141 Theory of Computation
  - Computability, complexity classes, languages and automata
- COMP6741 Algorithms for Intractable Problems
  - Preprocessing, approximation, parameterized, exponential-time, quantum, and heuristic algorithms
- Various MATH courses also build on the content covered in this course.

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Portfolio

- Tasks available on Formatif
- In each task, incorporate lab instructor's feedback until task complete
- Contract grading
  - Choose your target grade
    - Be realistic; discuss with lab instructor in task 0.0 Introduce Yourself
    - You can change this at any time
  - Attempt tasks up to this level inclusive
- At end of term, complete the Create Portfolio task and include a Learning Summary Report to recap what you found instructive and highlight your attainment of learning outcomes

Portfolio Evaluation

- Portfolio is evaluated holistically by your lab instructor
  - Mostly on the basis of task completion (quantity of satisfactory work)

Grade level	PS	CR	DN	HD
Total weight	55	15	15	15

  - Some adjustment for quality (subjective)
    - Includes writing, demonstrations, partial progress on incomplete tasks and more.
  - Typically $\pm 5\%$ of raw portfolio mark calculated purely from quantity (aggregate of completed tasks)
- Weighted 60% of course mark

Formatif

- Task types: (D)emonstration, (M)oodle, (R)egular, (L)aTeX

Task type	(D)	(M)	(R)	(L)
Relative weight	10	2	5	2

- Task sheets on Formatif
- Task resources on Moodle, suitable for import to Overleaf
- Start with tasks 0.1 *Getting Started* and 0.2 *Preparing for Success*

Final Exam

- Multiple choice, short answer and algorithm design problems
- Easier than comparable portfolio tasks
- Hurdle: 40% required to pass the course
- INSPERA (on campus): more info here
- Weighted 40% of course mark

Bonus Marks

- Reward contributions to other students' learning, in classes and on the forum

- About 5% of students will get at least 1 mark, half as many for each extra mark up to a maximum of 5

Lectures

- Weeks 1–5, 7–10 (no lectures in flex week)
  - Monday week 4 cancelled due to Labour Day public holiday
- Face to face
  - WEB stream students are welcome to attend in person
- Recording and live stream on Echo360
- Slides on Moodle

Consultation

- Mon 13:00 – 14:00 and Wed 15:00 – 16:00 at K17-202
- Weeks 1–5, 7–10 (no consultation or in flex week)
  - Monday week 4 cancelled due to Labour Day public holiday
- For questions about lecture content. Questions about tasks should be asked during labs and help sessions.
- Exam consultation: TBA

Seminar - tutor presents worked examples

- Weeks 1–5, 7–10 (no seminar in flex week)
- Reinforcement: revise recent lecture content and demonstrate problem solving process
- Read the tutorial sheet in advance to get the most benefit
- Online seminar is recorded

Labs - demonstrators support you with assessments

- Weeks 1–5, 7–10 (no labs in flex week)
- Free time to:
  - work on new Formatif tasks
  - clarify feedback on existing tasks
  - get help from the lab demonstrators
  - task discussions
  - get marked off for completed tasks
  - lab demonstrators will spend a max. of 10 minutes for discussions with one student per lab

Support

- In addition to classes, we provide the following support services and resources:
  - Discourse link
  - Drop-in help sessions starting in Week 1:
    - Mon 9am–11am in Quad G027
    - Tue 2pm–4pm in CSE Help (K17-G05)
    - Thu 6pm–8pm in CSE Help (K17-G05)

Equitable Learning Services

- Equitable Learning Services is a free and confidential service that provides practical support to ensure that your health condition doesn't adversely affect your studies.

- If you have an Equitable Learning Plan, please provide it to us by email within the first two weeks of the term.

How We Help You at UNSW

General Health and Wellbeing	Educational Adjustments
Feelings and Mental Health	Academic and Study Skills
Sexual Health & Relationships	Special Consideration
Uni and Life Pressures	Aboriginal & Torres Strait Islander Community
Reporting Harassment and Gendered Violence	Legal & Advocacy Services

Feedback

This course is evaluated each session using the myExperience system.

- Changes from last term:
  - In the previous offering, students reported that the lectures were sometimes behind schedule.
  - The lecture schedule has been rearranged to cover critical topics earlier in the term and efforts will be made to keep to the published schedule as closely as possible.
- Feedback is always welcome, e.g.
  - myExperience survey
  - feedback category in Discourse
  - email

That's All, Folks!!